To estimate the enthalpy of neutralization of hydrochloric acid and sodium hydroxide

Video Reference :- https://www.youtube.com/watch?v=ELSu7zmVRMg

Materials Required :-

- Laboratory Stand
- Calorimeter
- Cotton Wool
- Stirrer
- Beaker 250 ml
- Beaker 500 ml
- Measuring Cylinder 100 ml
- 1M HCL 100 ml
- 1M NaOH 100ml
- Thermometer

Procedure :-

- Make a setup of a calorimeter, take an expanded wooden box & fill it with cotton.
- Transfer 100ml of 1 molar hydrochloric into the beaker using the measuring cylinder.
- Place the hydrochloric acid into the calorimeter.
- Measure the initial temperature of the hydrochloric acid & record the data T 1°C.
- Now transfer 100 ml of 1 Molar Sodium Hydroxide solution in 250 ml Beaker using the measuring cylinder.
- Now measure the initial temperature of the base Sodium Hydroxide & record the data same as T 1℃ .
- Wait for some time for both solutions to attain the constant temperature (28 ℃).
- Now pour the Sodium Hydroxide solution into the calorimeter that contains 100 ml of HCL.
- Place the stirrer inside the calorimeter, close it & stir the mixture.
- Measure the maximum temperature of the reaction mixture & note it as the final temperature T 2℃

Observation :-

The Initial temperature of the acid & base ($T1\ ^\circ C$)	$28\ ^\circ C$
The final temperature after neutralization ($T2\ ^\circ C$)	$40\ ^\circ C$
Change in temperature t	$(t1-t2)\ ^\circ C$
Mass of the mixture solution after neutralization	200 g
The water equivalent of a calorimeter	Wg

Calculation :-

- Enthalpy change during neutralization of 100 ml of 1M HCL = (200 x W) x (t1 - t2) x 4.18
- Where W is the calorimeter constant of the value of 29.1 J (value depends on tha calorimeter)
- When 1000mL of 1M HCL is allowed to neutralize 100mL of 1M NaOH, calculate the heat that is released. This amount would be ten times greater than what was achieved.
- So, Enthalpy of neutralization of strong acid & base
  =(200 x W) x (t1 - t2) x 4.18 / 1.0 x 1000 / 100 J
  =(200 x W) x (t1 - t2) x 4.18 / 1.0 x 100 kJ
  =(200 X 29.1) X (28 - 40) x 4.18 / 1.0 x 100 kJ
  Enthalpy of neutralization of strong acid & base = **2919.32 kJ**

Result :-

- Enthalpy change in neutralizing strong acid (HCl) with base (NaOH) is **2919 kJmol**